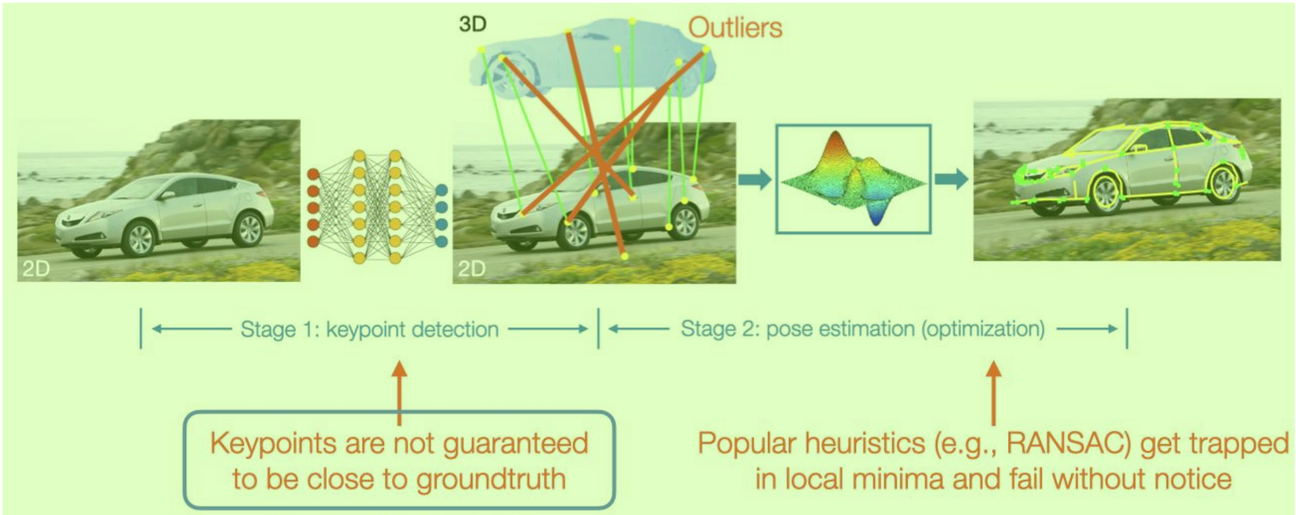
Conformal Semantic Keypoint Detection with Statistical Guarantees

Heng Yang and Marco Pavone [1]

Workshop on Robot Learning: Trustworthy Robotics [2]

Object Pose Estimation: why untrustworthy? [3]



[4]

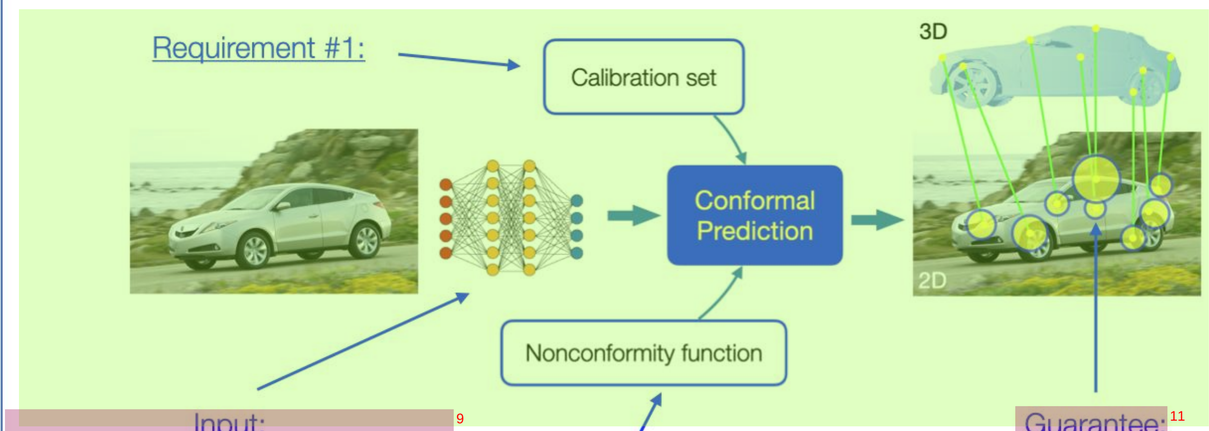
Focus of our paper

Related work: certifiable algorithms [YC22TPAMI]

Our method: conformal keypoint detection [5]

Paradigm shift: from heuristic prediction to probabilistically correct prediction sets [6]

Key ingredient: inductive conformal prediction (ICP) — finite-sample uncertainty quantification without assumptions on the data distribution or the prediction model [7]



[8]

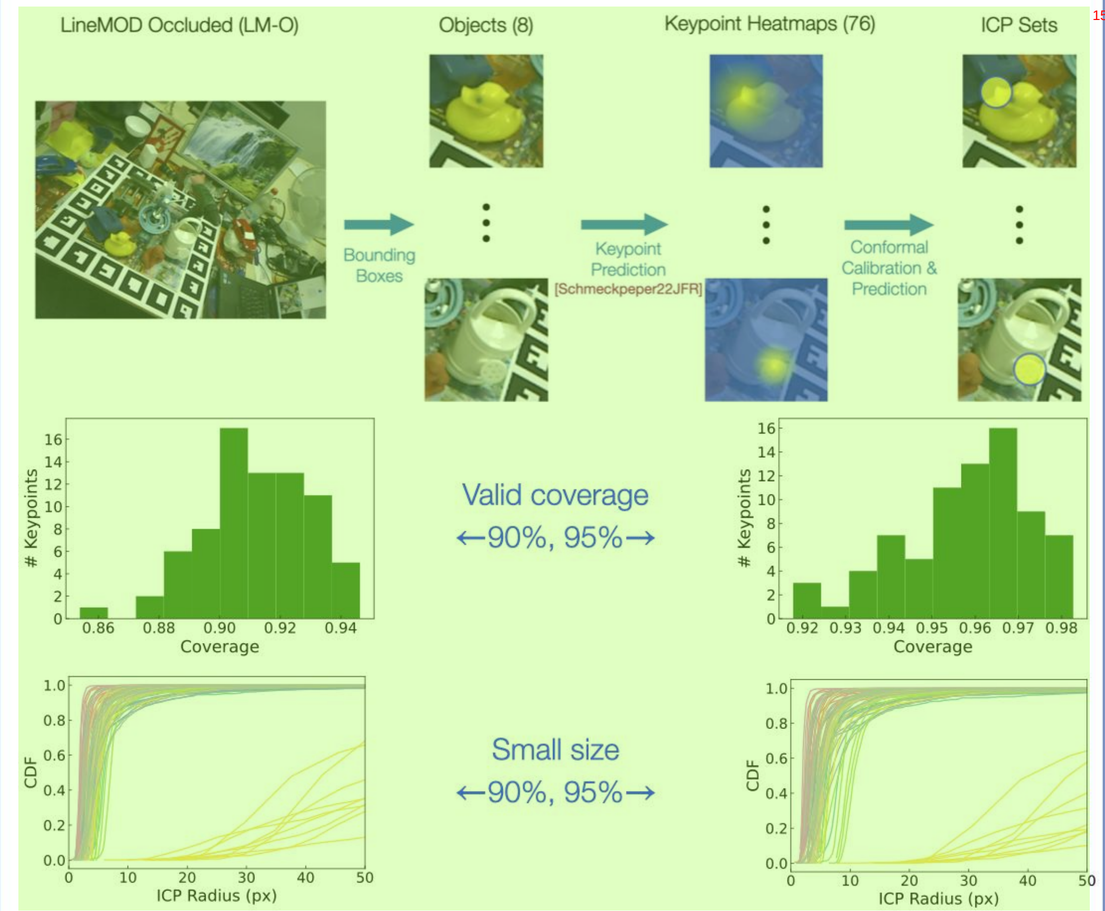
Input: [9]
Heuristic prediction (e.g., heatmap of the keypoint location)

Requirement #2: [10]
See paper for details!

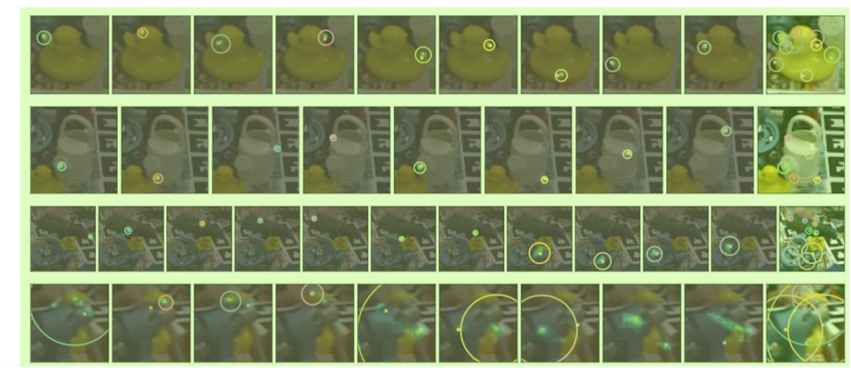
Guarantee: [11]
Cover the true keypoints with a user-specified probability (e.g., 90%) [12]

Experiments: works out of the box! [13]

Setup: 200 images for calibration, 1214 images for testing [14]
User: give me 90% and 95% coverage of the true keypoints



[15]



[16]